INSIDE STUDIO SP · VST3 · AU

# LIMÃO

ACID BASSLINE

## Step-by-step manual

Version 0.6.2 · Windows and macOS · VST3 and AU




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# What is LIMÃO

LIMÃO is a monophonic **acid bassline** synth with the soul of acid house and acid techno lines, but modern. It has:

- **One voice** with a SAW or SQUARE oscillator and an 18 dB/octave ladder filter that "screams" at high resonance.
- **ACCENT**, which adds level and opens the filter; consecutive accents stack up.
- **SLIDE**, which glides the pitch between tied notes.
- **A step sequencer of up to 32 steps**, synced to the project tempo, with scales, automatic randomization and MIDI export of the line.
- **Output effects**: DRIVE (TUBE, PEDAL, BITCRUSH), ping-pong DELAY and REVERB.
- **360 factory presets** in 9 categories, each with its own sequence pattern.

## Before you start

| ITEM | DETAILS |
|---|---|
| Format | VST3 instrument (Windows and macOS) and AU (macOS). |
| Voices | One (monophonic), with last-note priority. |
| Clock | The sequencer follows the project tempo and only runs while the DAW is playing. |
| Window | From 75% to 200%, chosen in the preset menu and saved with the project. |

## Installation and activation

1. Download the installer from the link in your purchase e-mail and choose Windows or Mac.
2. Run the installer and open your DAW. If needed, have it scan for new plugins (rescan).
3. Load LIMÃO on an **instrument track** (MIDI).
4. The first time, enter the **e-mail** and the **license token** you received.

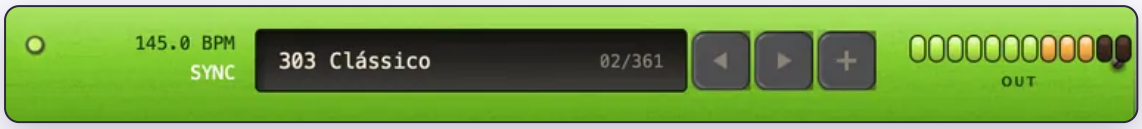
> **LICENSE.** The same token works on up to 2 computers. Deactivating it on one computer frees the slot for another. To view the license: click the preset name › `Licença…` (License…).

> **UNTIL ACTIVATED**, the instrument stays silent.

> **NEW VERSION.** With an active license, the bottom of the window tells you when a new version is out. `ATUALIZAR` (update) downloads and installs it on its own (on Windows it waits for you to close the DAW; on Mac it asks for your password). `BAIXAR` (download) opens the download in your browser.

STEP 1

# 1 Pick a preset




Top bar: BPM and clock status, preset name, ◀ ▶ to browse and + to save, output meter (OUT).

- Click the **preset name** to open the menu: bank › category › preset.
- Use ◀ ▶ to step through the list.
- There are 360 presets in 9 categories: *CLÁSSICO* (classic), *ACID HOUSE, ACID TECHNO, TRANCE, BASS, LEAD, SQUELCH, DUB* and *FX*, plus Init.

> **TIP.** Every preset brings the sound **and** a sequence pattern. It's the fastest way to hear LIMÃO: pick a preset and hit play in the DAW. Changing presets doesn't turn the sequencer on or off.

STEP 2

# 2 Play: sequencer or keyboard

The `SEQUENCER` button decides who plays the notes. It's on by default.

| MODE | HOW IT WORKS |
|---|---|
| **SEQUENCER on** | The step pattern plays in the project tempo while the DAW is playing. The key you hold on your keyboard **transposes the pattern** (C3 = no transposition). |
| **SEQUENCER off** | LIMÃO plays from the keyboard (MIDI), one note at a time, with last-note priority. Overlapping (legato) notes slide, and velocity 100 and above becomes an accent. |

Next to the BPM, the top bar shows the clock status:

- `SYNC`: the DAW is playing and the sequencer runs along with it.
- `AGUARDANDO PLAY` (waiting for play): the sequencer is on, but the DAW is stopped. Hit play.
- `LIVRE` (free): the sequencer is off; LIMÃO plays from the keyboard.

STEP 3

# 3 Build the line in the sequencer




Sequencer: controls on the top row and steps 1 to 16 (steps 17 to 32 are on the row below).

## Each step

| PART OF THE STEP | WHAT IT DOES |
|---|---|
| Note display | **Drag vertically** or use the **mouse wheel** to change the note (C1 to C4). **Double-click** toggles between note and rest. |
| `NOTA` / `PAUSA` / `LIGA` | The step key switches between playing a note (NOTA), staying silent (PAUSA, rest) or **tying** (LIGA: holds the note of the previous step). |
| `ACC` | Accent: the note sounds louder with the filter more open (and a short decay). |
| `SLD` | Slide: the pitch glides into the note of the next step. |

## Sequencer controls

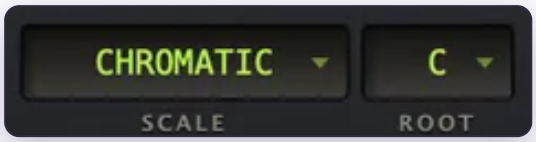
| CONTROL | RANGE / DEFAULT | WHAT IT DOES |
|---|---|---|
| **SEQUENCER** | on/off · **on** | Turns the sequencer on (see Step 2). |
| **PASSOS** (steps) | 1–32 · **16** | Length of the line. |
| **RATE** | 1/32, 1/16T, **1/16**, 1/8T, 1/8 | Length of each step. |
| **SWING** | 0–75% · **0%** | Adds swing to the groove of the line. |
| **GIRAR** (rotate) `<` `>` | — | Shifts the whole line one step to the left or right. |
| **TRANSP** `-1` `+1` | — | Moves all notes up or down (one scale degree, if a SCALE is set). |
| **ALEATORIZAR** (randomize) | — | Creates a new line instantly. |
| **LIMPAR** (clear) | — | Erases the line (every step becomes a rest). |

> **TIP.** The line you edit is saved with the project. To keep it together with the sound and use it in other projects, save it as a preset (Step 7).

STEP 4

# 4 Scale, randomization and MIDI

## SCALE + ROOT: everything in key




- Choose the scale in `SCALE` and the root in `ROOT` (C to B). There are 13 scales: MAJOR, MINOR, DORIAN, PHRYGIAN, LYDIAN, MIXOLYDIAN, LOCRIAN, HARM MINOR, MEL MINOR, PENTA MAJ, PENTA MIN, BLUES and PHRYG DOM.
- Every step is snapped to the scale: an out-of-scale note moves to the nearest note in the scale, and the display shows the note that actually plays.
- Dragging the note, the mouse wheel and TRANSP move **one scale degree** at a time.
- **CHROMATIC** turns the scale off. The stored line doesn't change: switch back to CHROMATIC and it plays exactly as it was.

## ALEATORIZAR and AUTO ALEATÓRIO

- `ALEATORIZAR` generates a new line once, using only scale notes, starting from the root.
- `AUTO ALEATÓRIO` (auto random, on the sequencer title row) keeps changing the line on its own, every `A CADA` (every) 4, 8, **16** (default), 32 or 64 beats. The change always lands on the bar line.
- The new lines **grow out of yours**: they stay in the scale and root, between the lowest and highest notes you used, and keep the same number of notes, rests, ties, accents and slides. Each change moves only 1 or 2 rhythm steps, so the sound evolves gradually.
- Editing the line, loading a preset, pressing ALEATORIZAR or changing PASSOS makes the next change start from the new line.

## ARRASTAR MIDI: take the line to your DAW

Click and **drag** the `ARRASTAR MIDI` (drag MIDI) key onto a DAW track: the line becomes a MIDI clip, already with SCALE + ROOT, accent (velocity 127), slide (note touching the next one), swing and tempo.

> **TIP.** Use AUTO ALEATÓRIO to find ideas and, when you like a line, turn AUTO off and drag the MIDI into your DAW.

STEP 5

# 5 Shape the sound: VOZ (voice) and MACRO




| CONTROL | RANGE / DEFAULT | WHAT IT DOES |
|---|---|---|
| **WAVE** | **SAW** / SQUARE | Oscillator waveform: saw (brighter) or square (hollower). |
| **TUNE** | ±12 st · **0.00 st** | Tuning, in semitones. |
| **CUTOFF** | 40 Hz–12 kHz · **520 Hz** | Filter opening: lower = darker. |
| **RESONANCE** | 0–100% · **65%** | Boost at the cutoff frequency; turned up, the filter "screams". |
| **ENV MOD** | 0–100% · **55%** | How much the envelope opens the filter on each note (the "wow"). |
| **DECAY** | 30 ms–2.5 s · **380 ms** | How long the filter takes to close after the note. |
| **ACCENT** | 0–100% · **60%** | Strength of the ACC steps: adds level and cutoff. Consecutive accents stack up. |
| **SLIDE** | 10–400 ms · **60 ms** | Glide time between SLD notes (or overlapping notes on the keyboard). |
| **ÁCIDO** (acid, MACRO) | 0–100% · **0%** | Opens ENV MOD, RESONANCE and a bit of CUTOFF together. One knob to make the line more acid. |

The **oscilloscope** on the right shows the outgoing waveform, the current CUTOFF and a bar whenever an ACCENT hits.

## Using the knobs

- **Drag** up and down, or use the **mouse wheel**.
- **Double-click** resets to the default value.

> **ACID RECIPE.** Start with a low CUTOFF, high RESONANCE and ENV MOD in the middle. Put ACC and SLD on a few steps and slowly turn CUTOFF (or ÁCIDO) while the line plays. Automate that move in your DAW.

STEP 6

# 6 Effects and output




| BLOCK | CONTROLS / DEFAULT | WHAT IT DOES |
|---|---|---|
| **DRIVE** | TIPO (type): **TUBE**, PEDAL, BITCRUSH · DRIVE 0–100% (**20%**) | TUBE is warm and asymmetric; PEDAL clips hard; BITCRUSH reduces bit depth and sample rate together. |
| **DELAY** | on/off (**off**) · TIME 1/16 to 3/8 (**3/16**) · FEEDBACK 0–95% (**45%**) · MIX (**25%**) | Tempo-synced ping-pong echo; each repeat gets darker. |
| **REVERB** | on/off (**off**) · SIZE (**45%**) · MIX (**20%**) | Ambience. |
| **SAÍDA** (output) | VOLUME −36 to +6 dB (**−6 dB**) | Final level. The LED meter (OUT) on the top bar shows the peak. |

> **VOLUME.** High RESONANCE, ACCENT and DRIVE all add level. If OUT lights up red, turn VOLUME down.

STEP 7

# 7 Save your own presets

- In the preset menu, `Salvar como novo preset…` (save as new preset, or the `+` button) stores **the sound and the pattern** in the **USER** list.
- Your presets can be overwritten, renamed and deleted (they go to the trash).
- `Importar banco de presets…` (import preset bank) installs a **.lmobank** bank, which appears below FACTORY 1. `Remover banco` (remove bank) sends the bank to the trash.
- The same menu holds the **window size** (75–200%), the license and the version.

| SYSTEM | USER PRESETS FOLDER |
|---|---|
| Windows | %APPDATA%\InsideStudioSP\LIMAO\Presets |
| macOS | ~/Library/Audio/Presets/InsideStudioSP/LIMAO |

# FAQ and troubleshooting

| SITUATION | WHAT TO DO |
|---|---|
| No sound | Check that the license is active (preset name › Licença…): until activated, the instrument stays silent. Also check VOLUME and that the track is receiving MIDI or the DAW is playing. |
| The top bar shows AGUARDANDO PLAY | The sequencer is on and only runs while the DAW is playing. Hit play on the project. |
| I play the keyboard and the line changes key | With the sequencer on, the held key transposes the pattern (C3 = no transposition). To play single notes, turn the SEQUENCER off. |
| The notes changed on their own | AUTO ALEATÓRIO is on. Turn it off to keep the line fixed. |
| A note doesn't play the pitch I entered | With a SCALE set, the note is snapped to the scale. The display shows the note that plays. Use CHROMATIC to play it as written. |
| Sound distorting or too loud | Lower DRIVE, RESONANCE or ACCENT, or the SAÍDA VOLUME. |
| The slide doesn't happen | SLD glides into the next step, which must be a note (not a rest). On the keyboard, play overlapping (legato) notes. |
| Window too big or too small | Preset menu › window size (75–200%). It's saved with the project. |

## 30-second summary

1. LIMÃO on an instrument track → activate the license.
2. Pick a preset and hit **play** in the DAW.
3. Edit the steps: note, NOTA / PAUSA / LIGA, **ACC** and **SLD**.
4. Set **SCALE + ROOT** to the song's key.
5. Turn **CUTOFF**, **RESONANCE**, **ENV MOD** or **ÁCIDO**.
6. Season with DRIVE, DELAY and REVERB.
7. Like it? **ARRASTAR MIDI** into the DAW or save it as a preset.

**SUPPORT.** Questions, suggestions or issues: **contato@insidestudiosp.com.br** · **insidestudiosp.com.br**